Q1: [10 marks]

a) **Discuss** two advantages that database systems have over conventional flat file systems.
Read in textbook

b) **Describe** the function of a precompiler in a DBMS.
Read in textbook

c) **Explain** the difference between a key and superkey with the help of an example.
Read in textbook

d) Consider the following relational database state

AIRPLANE_TYPE

typeid	Name
1	Boeing777
2	Airbus A380

AIRPLANE

plane_id	purchase_date	Num_seats	typeid
1	12-Dec-2008	400	1
2	01-Nov-2009	400	1
3	13-Nov-2010	800	2

Identify any foreign key present :
Typeid in airplane table refers to primary key of airplane type
and **discuss** all integrity constraints violated by each operation, if any.

i. Insert<null, 'Airbus 320'> into AIRPLANE_TYPE
Entity integrity constraint violated

ii. Update planeid to 2 where planeid is 3 in AIRPLANE
Key constraint violated

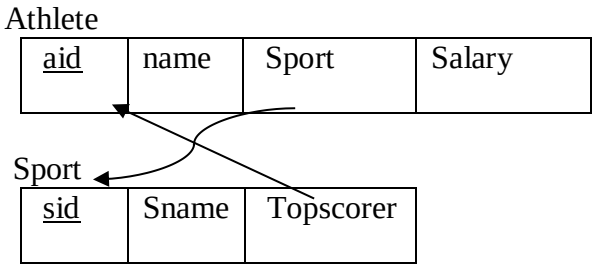
iii. Delete the AIRPLANE_TYPE tuple with typeid=2
Referential Integrity constraint violated

iv. Insert<4, '12-Oct-2025', 500, 2> into AIRPLANE
This will be successful

CLO # 2, PLO3, C6: Construct SQL statements given a relational data model to manage an electronic database.

Q2: [15 marks]

Please consider the relational data model below and **construct** appropriate SQL commands to answer the questions that follow



a) **Compose** the appropriate SQL (DDL) statements (in the order you would execute them in real-time on a DBMS) to define the database (with all the constraints).

Create table athlete (aid int primary key, name varchar(30), sport int, salary int)

Create table sport (sid int primary key, sname varchar(20), topscorer int ,
Constraint AFK foreign key (topscorer) references athlete (aid))

Alter table athlete add Constraint SFK foreign key (sport) references Sport (sid))

b) You want to ensure that nobody enters a negative value for salary column in the athlete table. **Compose** the SQL (DDL) statement to do this.

Alter table athlete add Constraint cSal check (salary>0)

c) You decide to get rid of table sport. **Compose** the DDL statement to do this.

Drop table sport

d) **Write** the SQL (DML) query to list the names of athletes whose name consist of "khan" alongwith the name of sport they participate in. Your output should be sorted on the basis of sport name.

Select a.name, s.sname
From athlete a inner join sport s on a.sport=s.sid
Where a.name like '%khan%'
Order by s.sname

e) **Write** the SQL (DML) query to list the names of sports which have a total number of athletes participating in it as more than 100.

```
Select  s.sname, count(*)
From athlete a inner join sport s on a.sport=s.sid
Group by  s.sname, s.sid
Having count(a.aid) >100
```